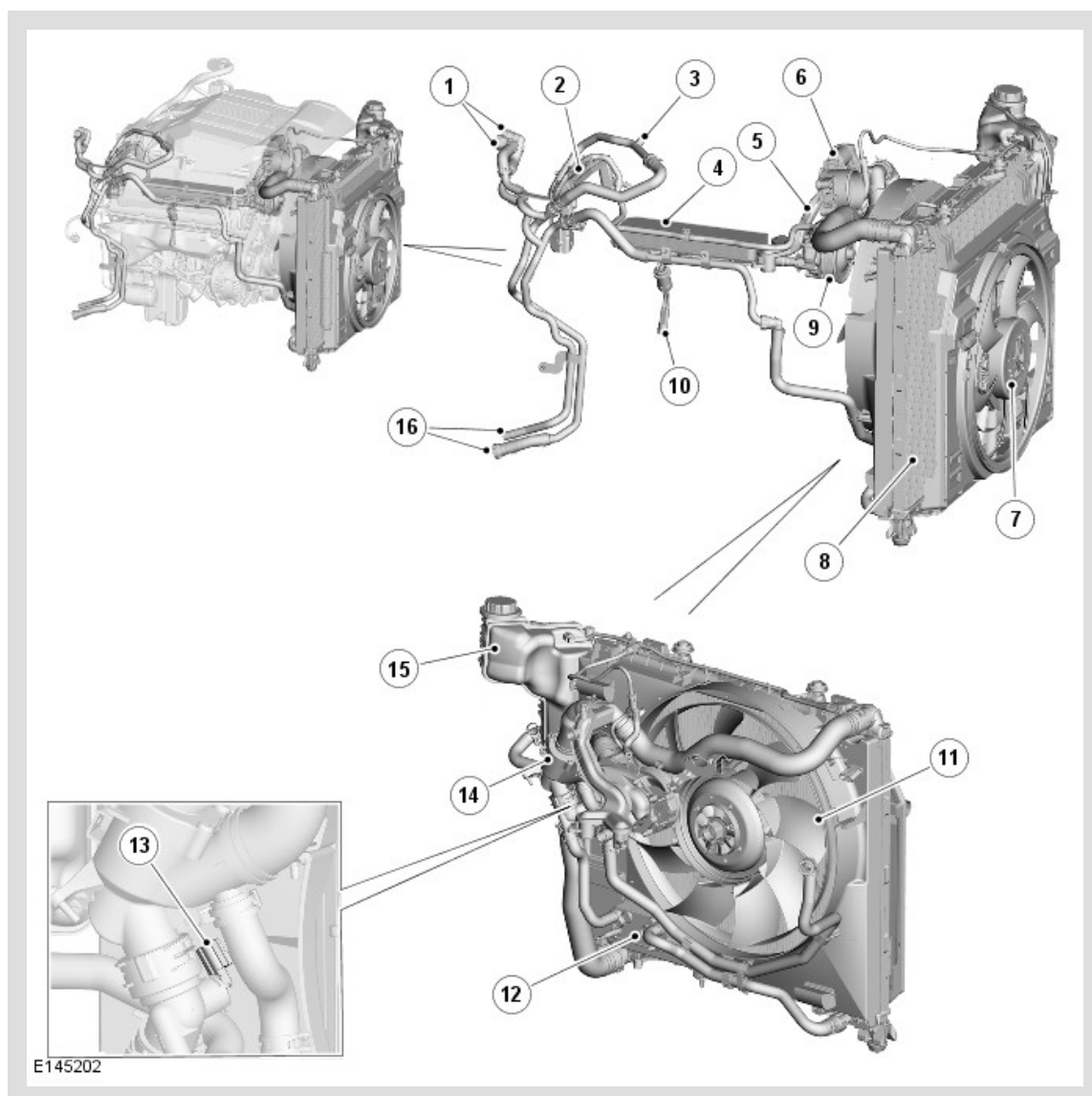


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2014.0 RANGE ROVER (LG), 303-03

ENGINE COOLING - V8 N/A 5.0L PETROL/V8 S/C 5.0L PETROL

COMPONENT LOCATION - NATURALLY ASPIRATED VEHICLES

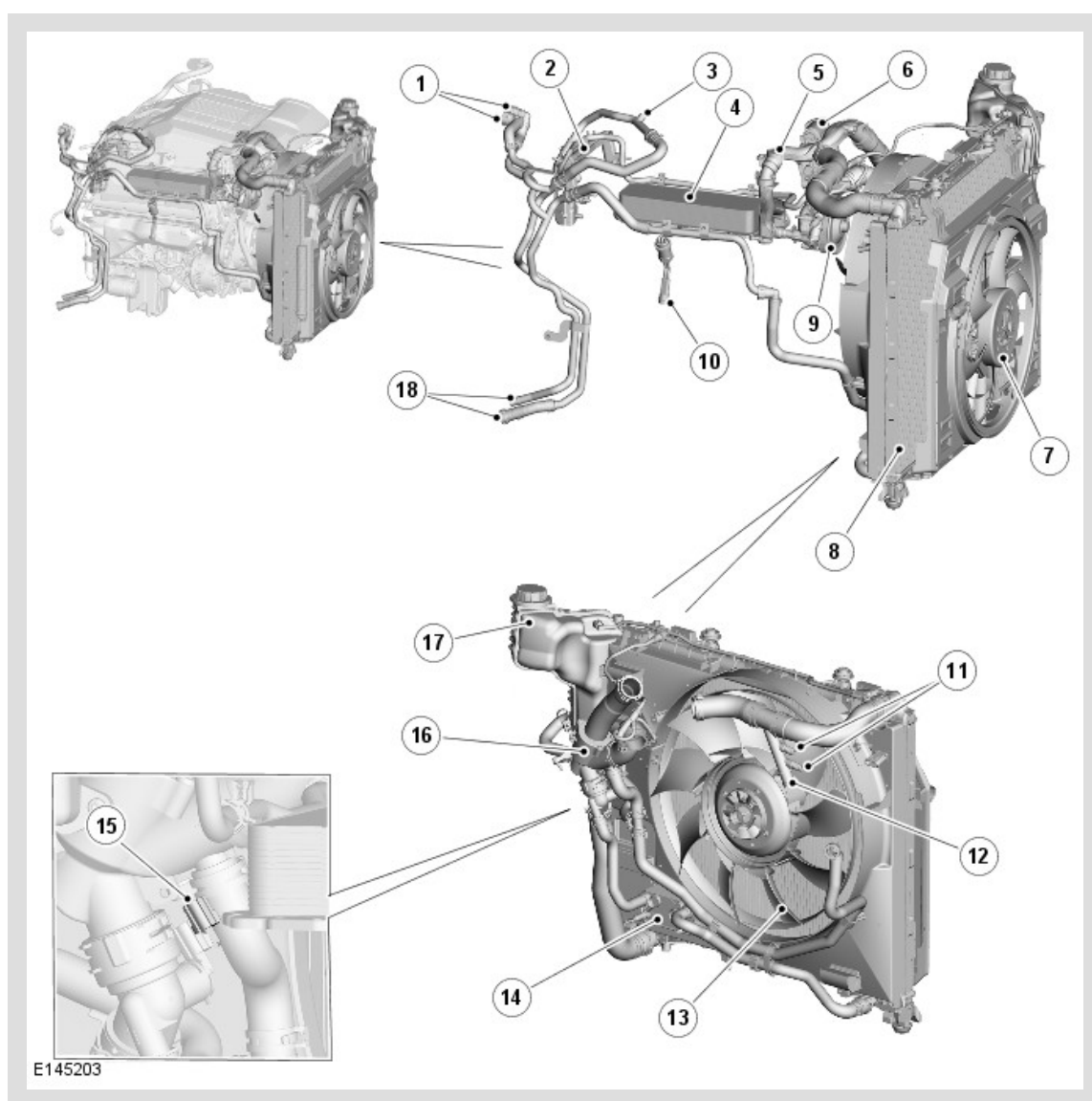


ITEM	DESCRIPTION
1	Heater connections
2	Heater manifold
3	Bleed screw
4	Engine oil cooler
5	Outlet tube
6	Electric throttle

7	Electric cooling fan (where fitted)
8	Radiator
9	Coolant pump
10	Block heater (where fitted)
11	Viscous cooling fan
12	Transmission oil cooler
13	Radiator outlet temperature sensor
14	Thermostat
15	Expansion tank
16	Auxiliary heater connections (where fitted)



COMPONENT LOCATION - SUPERCHARGER VEHICLES



ITEM	DESCRIPTION
1	Heater connections
2	Heater manifold
3	Bleed screw
4	Engine oil cooler
5	Outlet tube
6	Electric throttle
7	Electric cooling fan (where fitted)
8	Radiator
9	Coolant pump
10	Block heater (where fitted)

11	Connections for supercharger charge air coolers
12	Connection for supercharger charge air radiators
13	Viscous cooling fan
14	Transmission oil cooler
15	Radiator outlet temperature sensor
16	Thermostat
17	Expansion tank
18	Auxiliary heater connections (where fitted)

OVERVIEW

The engine cooling system maintains the engine within an optimum temperature range under changing ambient and engine operating conditions. The system is a pressurized expansion tank system with continuous bleeds to separate air from the coolant and prevent the formation of air locks. The engine cooling system also provides:

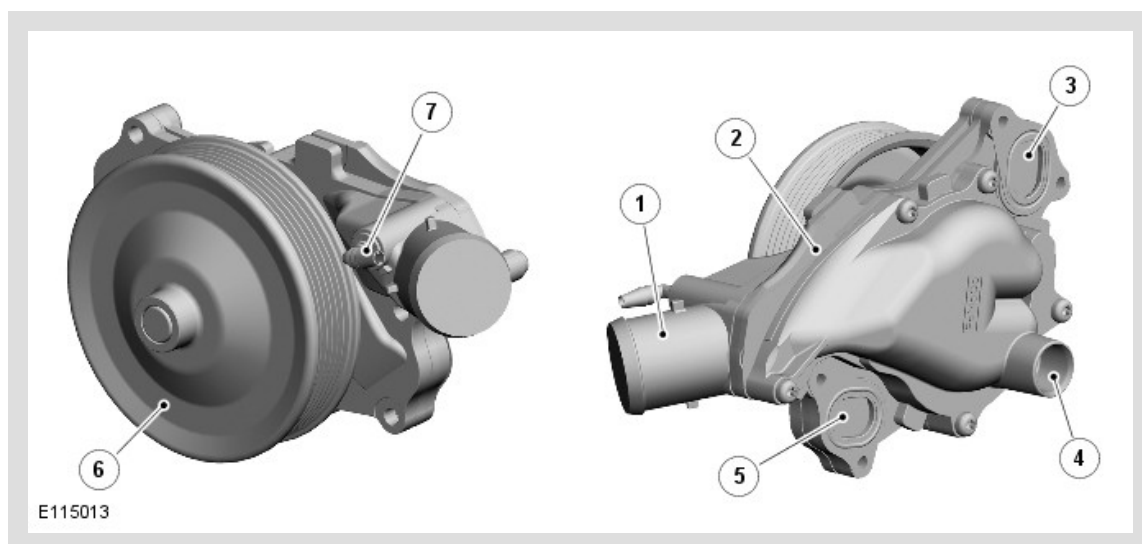
- Heating for:
 - The passenger compartment.
 - The throttle body.
- Cooling for:
 - The engine oil cooler.
 - The transmission fluid cooler.

The primary components of the engine cooling system are the:

- Coolant pump.
- Thermostat.
- Radiator.
- One or two engine cooling fans.
- Expansion tank.
- Outlet tube and heater manifold.
- Connecting hoses and pipes.

DESCRIPTION

COOLANT PUMP



ITEM	DESCRIPTION
1	Inlet connection
2	Pump body
3	Outlet flange to right cylinder head
4	Outlet to engine oil cooler
5	Outlet flange to left cylinder head
6	Pulley
7	Bleed pipe connection (containing check valve)

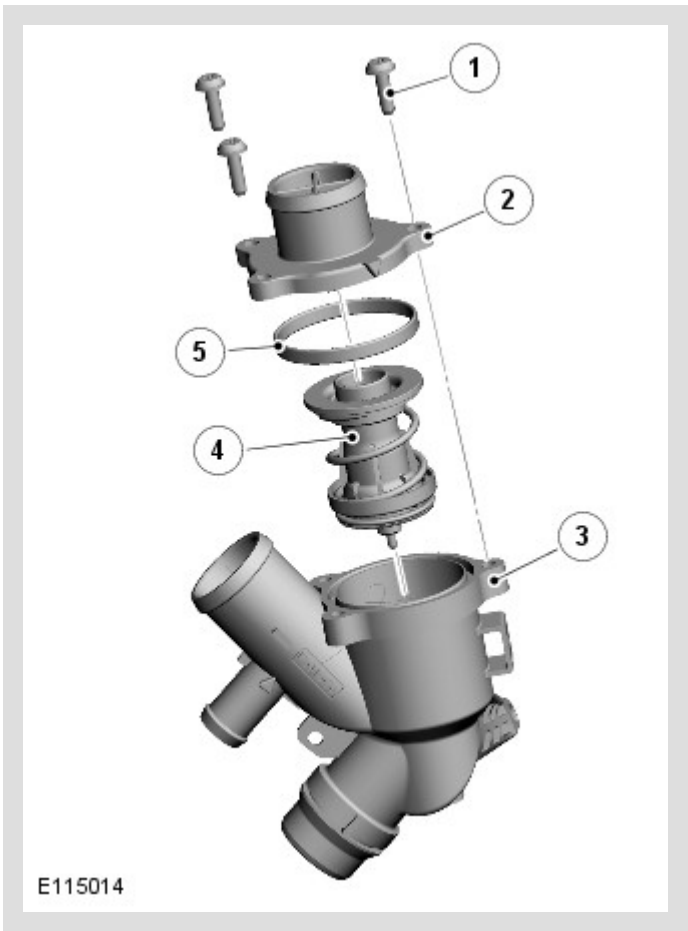
The body of the coolant pump contains an impeller attached to a shaft supported in a bearing assembly. The impeller is driven by a pulley, pressed on to the front of the shaft, which is driven by the accessory drive belt.

For additional information, refer to: Accessory Drive (303-05 Accessory Drive - V8 5.0L Petrol/V8 S/C 5.0L Petrol, Description and Operation).

Two coolant outlet flanges attach the coolant pump to the front of the cylinder block. A pipe connects a further coolant outlet to a pipe from the engine oil cooler. A bleed

connector is installed in the front of the coolant pump, adjacent to the coolant inlet connection from the thermostat. A check valve is incorporated into the bleed connection.

THERMOSTAT

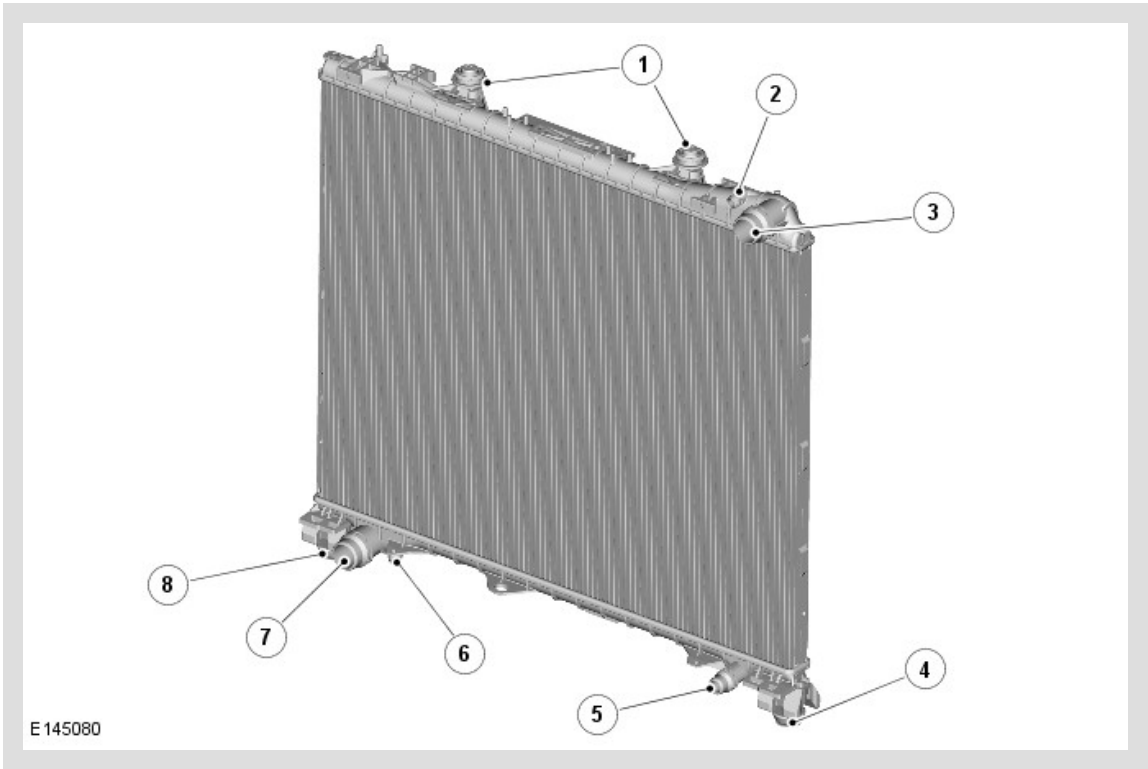


ITEM	DESCRIPTION
1	Screw (3 off)
2	Lower body
3	Upper body
4	Thermostat
5	Seal

The thermostat is a multi-stage device located in the coolant pump inlet to provide fast response and control of the engine outlet temperature.

The thermostat allows rapid engine warm-up by preventing coolant flow through the radiator and by limiting coolant flow through the cylinder block when the engine is cold. During warm-up and at engines speeds above approximately 1800 rev/min, a by-pass valve opens to control the coolant flow and pressure, to protect the engine components. When the thermostat opening reaches 6 mm (0.24 in.), the by-pass flow is shut-off. When the thermostat opening exceeds 6 mm (0.24 in.), the radiator coolant flow is further controlled up to the point where the thermostat is fully open. At this point maximum radiator coolant flow is achieved to provide maximum cooling.

RADIATOR



ITEM	DESCRIPTION
1	Upper mounts
2	Vent hose connection
3	Upper hose connection
4	Lower mount
5	Transmission hose connection
6	Drain screw

7	Lower hose connection
8	Lower mount

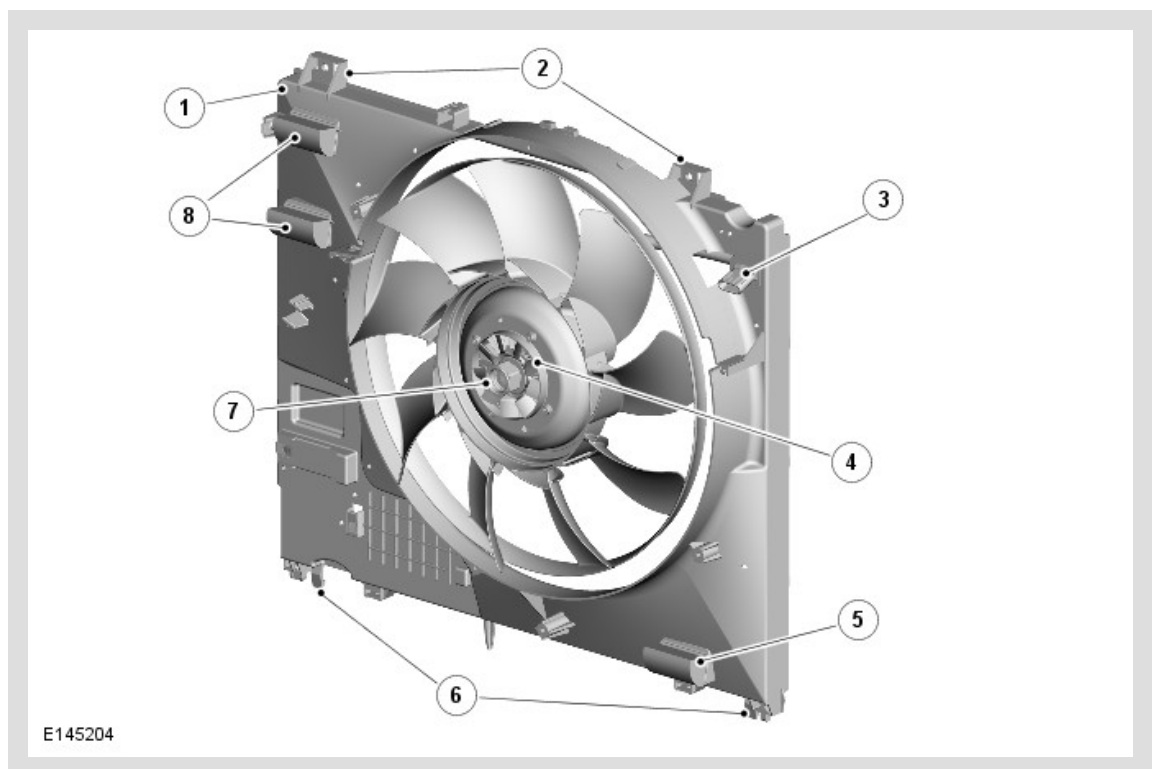
The radiator is a vertical-flow type with an integral transmission fluid sub-cooler. The radiator is located in the vehicle by bushes attached to the end tanks. The lower bushes are installed in the front subframe. The upper bushes are installed in the front end carrier.

Upper and lower hose connections are incorporated into the upper and lower end tanks respectively. The lower end tank also incorporates a drain screw, and a connection for the feed hose of the transmission fluid cooler. The upper end tank incorporates a connection for a vent hose to the expansion tank. The end tanks also incorporate mounting points for viscous and electric cooling fan shrouds, and the climate control condenser.

ENGINE COOLING FAN(S)

Depending on market and model specification, vehicles have either a viscous cooling fan, or a viscous cooling fan and an electric cooling fan, to help regulate the coolant temperature.

VISCOUS COOLING FAN



ITEM	DESCRIPTION
1	Shroud
2	Mounting lugs

3	Electrical connector
4	Electro-viscous drive unit
5	Speed flap
6	Mounting lugs
7	Securing nut
8	Speed flaps

The viscous engine cooling fan functions as a normal viscous fan, but with electronic control over the level of engagement of the viscous clutch. The ECM (engine control module) controls the level of engagement to optimize fan speed for all operating conditions.

The blades of the cooling fan are located in a shroud attached to the rear of the radiator. The shroud incorporates speed flaps, which pivot open to allow engine compartment cooling at speed. A securing nut attaches the cooling fan to a drive pulley, which is mounted on the front of the engine and driven by the accessory drive system.

For additional information, refer to: Accessory Drive (303-05 Accessory Drive - V8 5.0L Petrol/V8 S/C 5.0L Petrol, Description and Operation).

NOTE:

On naturally aspirated engines the securing nut has a LH (left-hand) thread; on SC (supercharger) engines the securing nut has a RH (right-hand) thread.

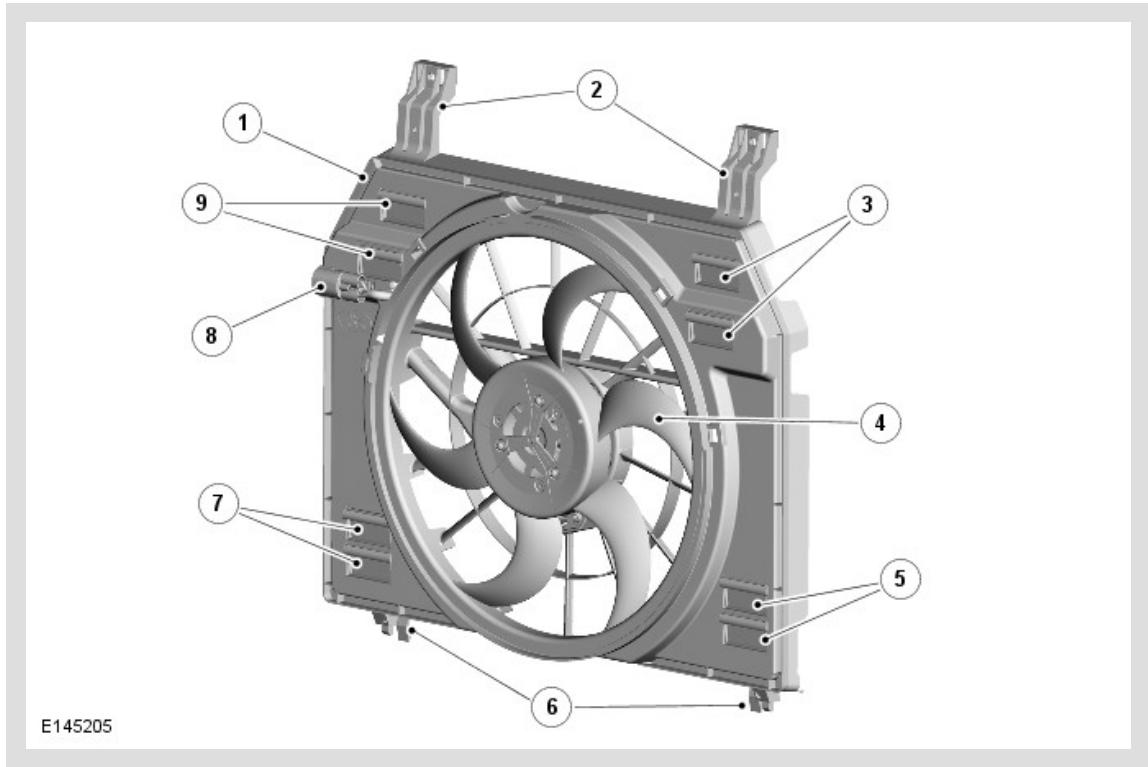
The viscous engine cooling fan is operated by a control module, integrated into the drive unit, which is controlled by the ECM . An electrical connector in the top right corner of the shroud provides the interface between the cooling fan harness and the vehicle wiring. The control module is provided with:

- Two battery power supplies from the ECM relay in the EJB (engine junction box)
- A PWM (pulse width modulation) signal from the ECM
- A ground.

A Hall effect sensor in the fan control module provides a fan speed signal for the ECM .

To control the cooling fan, the ECM varies the duty cycle of the PWM signal. The fan control module translates the duty cycle into a fan target speed and sets the level of engagement of the viscous clutch accordingly.

ELECTRIC COOLING FAN



ITEM	DESCRIPTION
1	Shroud
2	Mounting lugs
3	Speed flaps
4	Fan blades and motor
5	Speed flaps
6	Mounting lugs
7	Speed flaps
8	Electrical connector
9	Speed flaps

The electric engine cooling fan is a variable speed fan installed in a shroud attached to the front of the radiator. The shroud incorporates speed flaps, which pivot open to allow engine compartment cooling at speed.

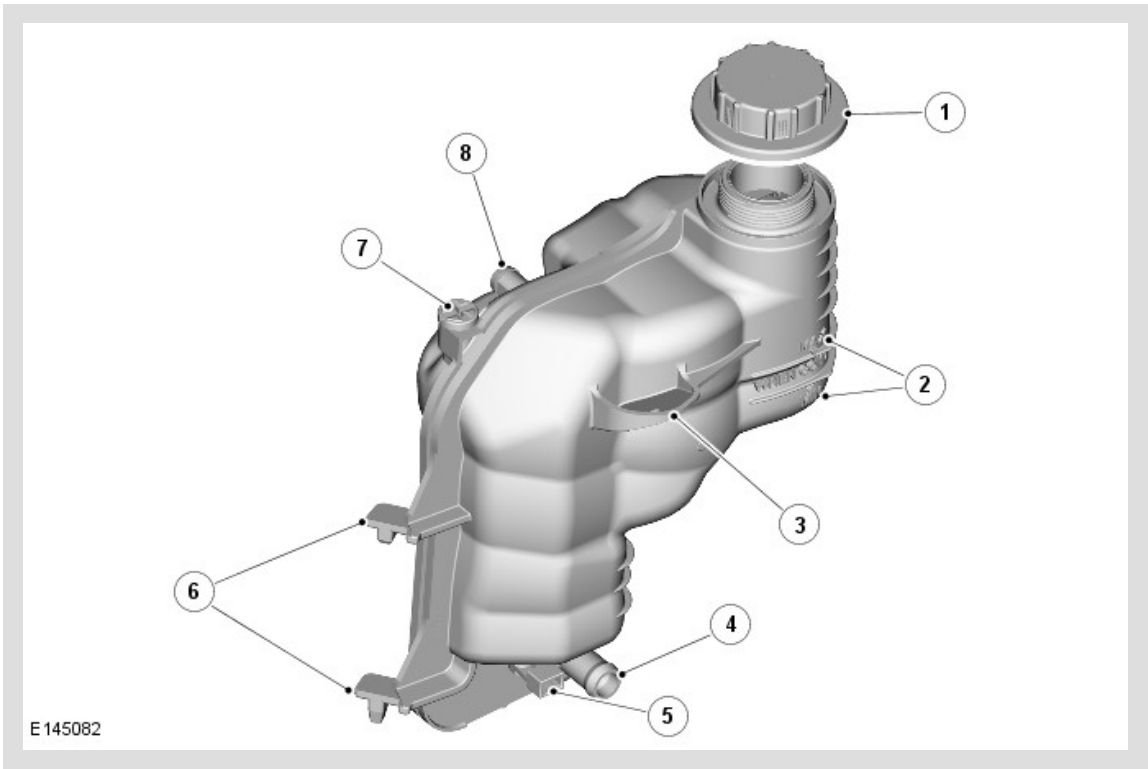
The cooling fan is operated by a fan control module, integrated into the electric motor, which is controlled by the ECM . An electrical connector at the right side of the fan shroud provide the interface between the cooling fan harnesses and the vehicle wiring.

The fan control module of each cooling fan is provided with:

- A battery power supply from the EJB
- An ignition signal from the ECM relay in the EJB
- A PWM signal from the ECM
- A ground.

To control the cooling fan, the ECM varies the duty cycle of the PWM signal. The fan control module translates the duty cycle into a fan target speed and sets the output current to the fan motor accordingly.

EXPANSION TANK



ITEM	DESCRIPTION
1	Filler cap
2	MAX and MIN level markings
3	Mounting lug

4	Supply hose connection
5	Coolant level sensor
6	Mounting lugs
7	Bleed screw
8	Vent hose connection

A pressurized expansion tank system is used which continuously separates the air from the cooling system and replenishes the system through a hose connected between the expansion tank and the thermostat. A continuous vent into the expansion tank, through a hose connected to the radiator, prevents air locks from forming in the cooling system.

The expansion tank is attached to the front end carrier in the front left corner of the engine compartment. A filler cap, bleed screw and level sensor are incorporated into the expansion tank. MAX and MIN level markings are molded into the exterior of the tank.

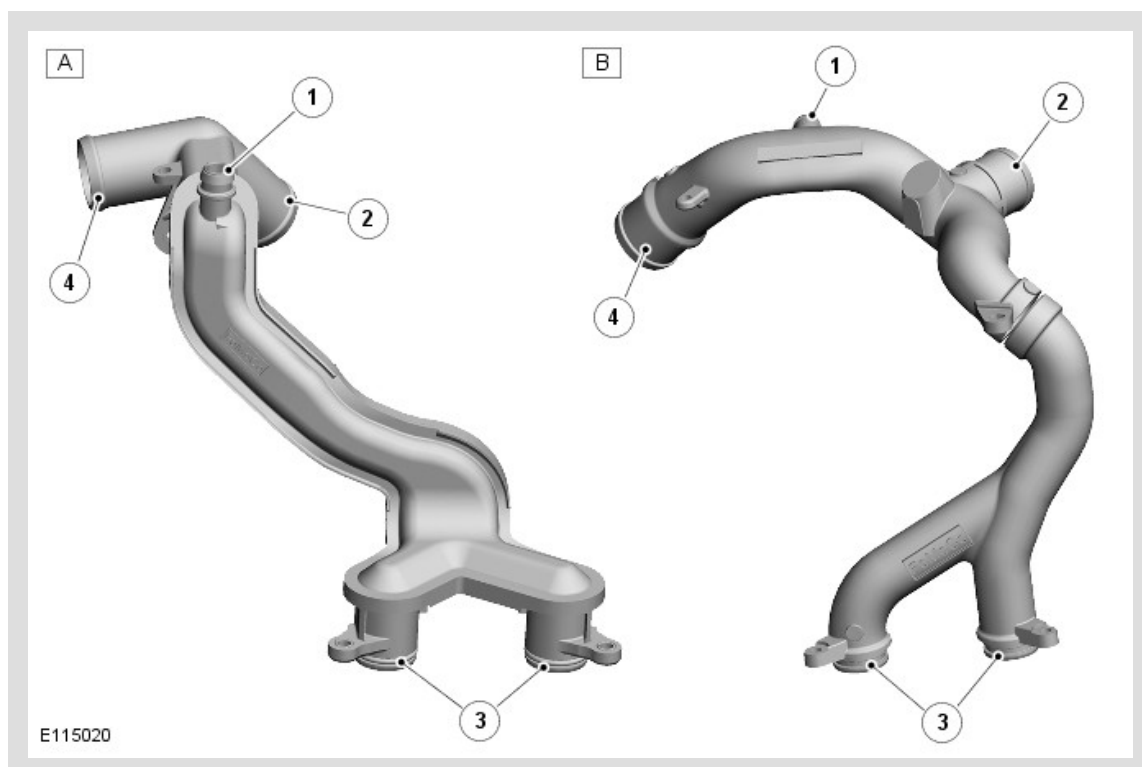
The expansion tank provides the following functions:

- Service fill
- Coolant expansion during warm-up
- Air separation during operation
- System pressurization by the filler cap

The expansion tank has an air space of approximately 0.5 to 1 liter (1.06 to 2,11 US pints), above the MAX level, to allow for coolant expansion.

OUTLET TUBE AND HEATER MANIFOLD

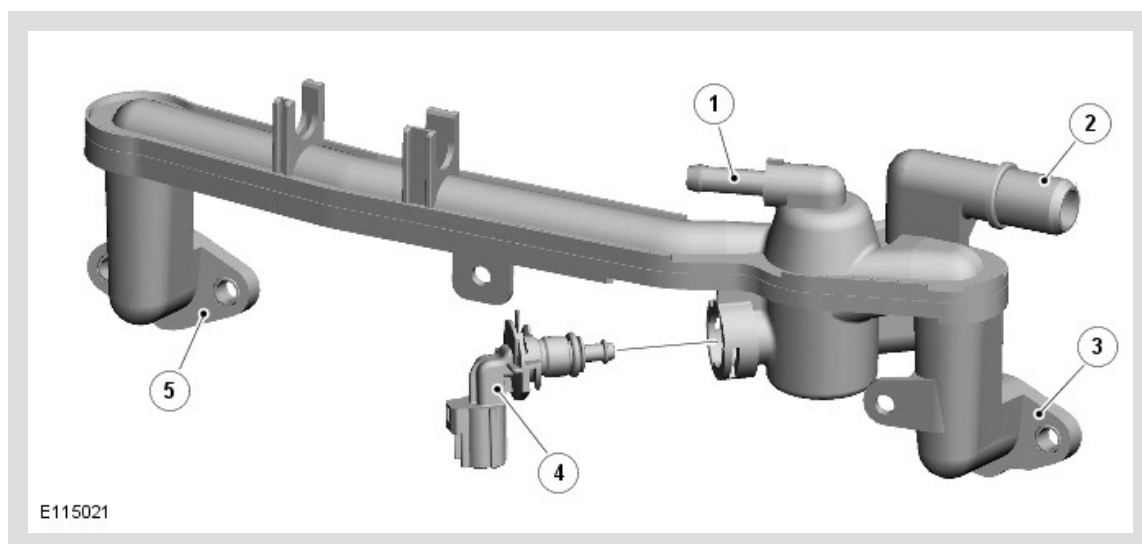
OUTLET TUBE



A = OUTLET TUBE ON NATURALLY ASPIRATED VEHICLES; B = OUTLET TUBE ON SUPERCHARGER VEHICLES.

ITEM	DESCRIPTION
1	Bleed spigot (fitted with blanking plug)
2	Radiator upper hose connection
3	Cylinder block connections
4	Thermostat hose connection

HEATER MANIFOLD



ITEM	DESCRIPTION
1	Throttle body heater hose connection
2	Heater core supply hose connection
3	Right cylinder head connection
4	Engine coolant temperature sensor
5	Left cylinder head connection

ENGINE COOLANT

The coolant is silicate free and must not be mixed with conventional engine coolant.

OPERATION

COOLANT CIRCUIT FLOW

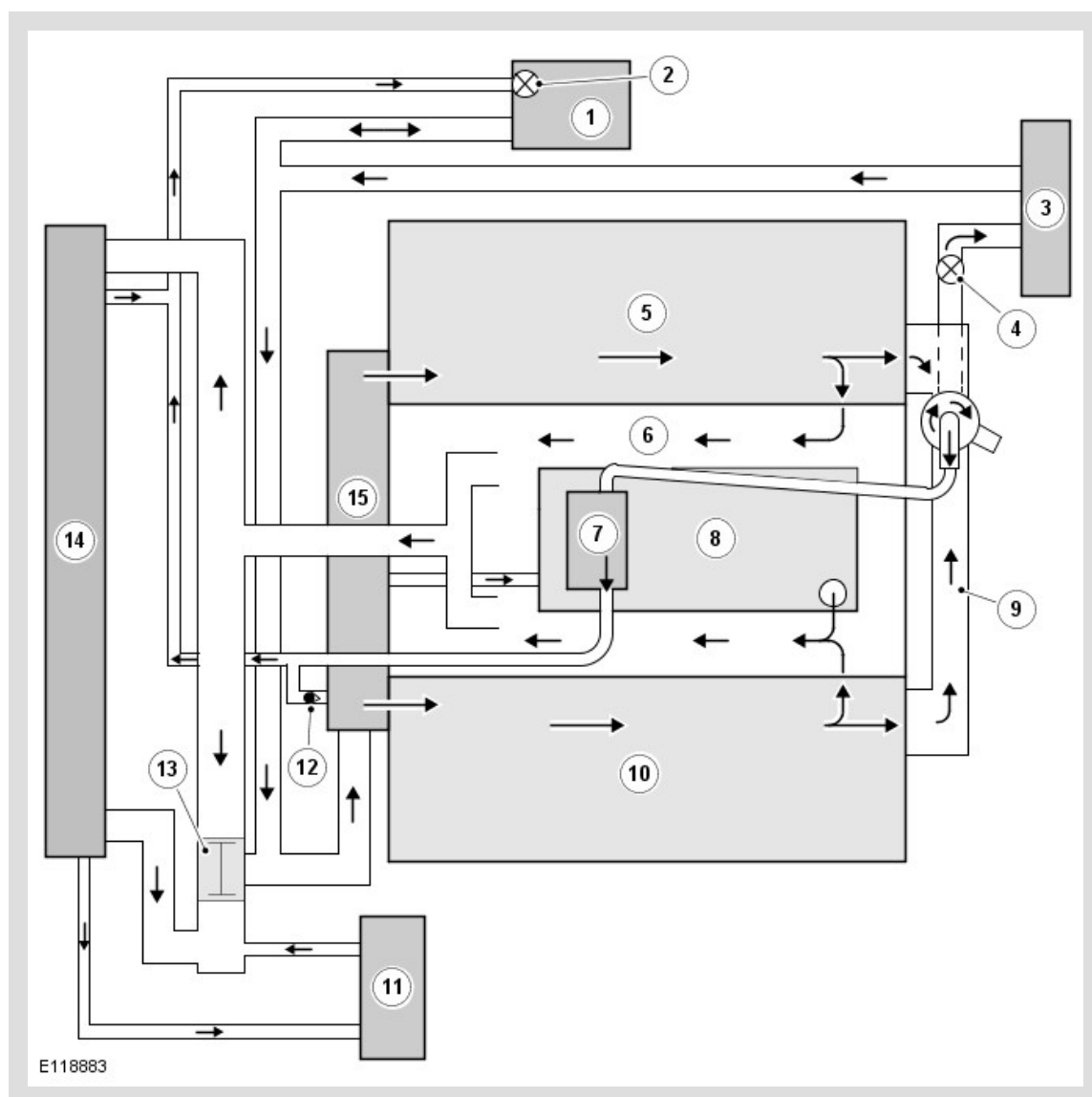
When the engine is running, the coolant is circulated around the engine cooling system by the coolant pump. From the coolant pump, coolant flows through the cylinder heads and the engine oil cooler into the cylinder block and the heater manifold.

In the cylinder block, the coolant flows forwards to the outlet tube. When the coolant is cold, the thermostat is closed and the coolant flows direct from the outlet tube back to the coolant pump. Once the coolant reaches operating temperature the thermostat begins to open, to control system temperature, and coolant flows from the outlet tube to the coolant pump via the radiator. When the thermostat is open, the coolant flow through the radiator also generates a coolant flow through the transmission fluid cooler.

From the heater manifold the coolant flows through the electric throttle and the heater core, in parallel circuits that are unaffected by the position of the thermostat. From the electric throttle, the coolant merges with bleed coolant from the coolant pump and the outlet tube and flows to the expansion tank. From the heater core, the coolant flows back to the inlet of the coolant pump.

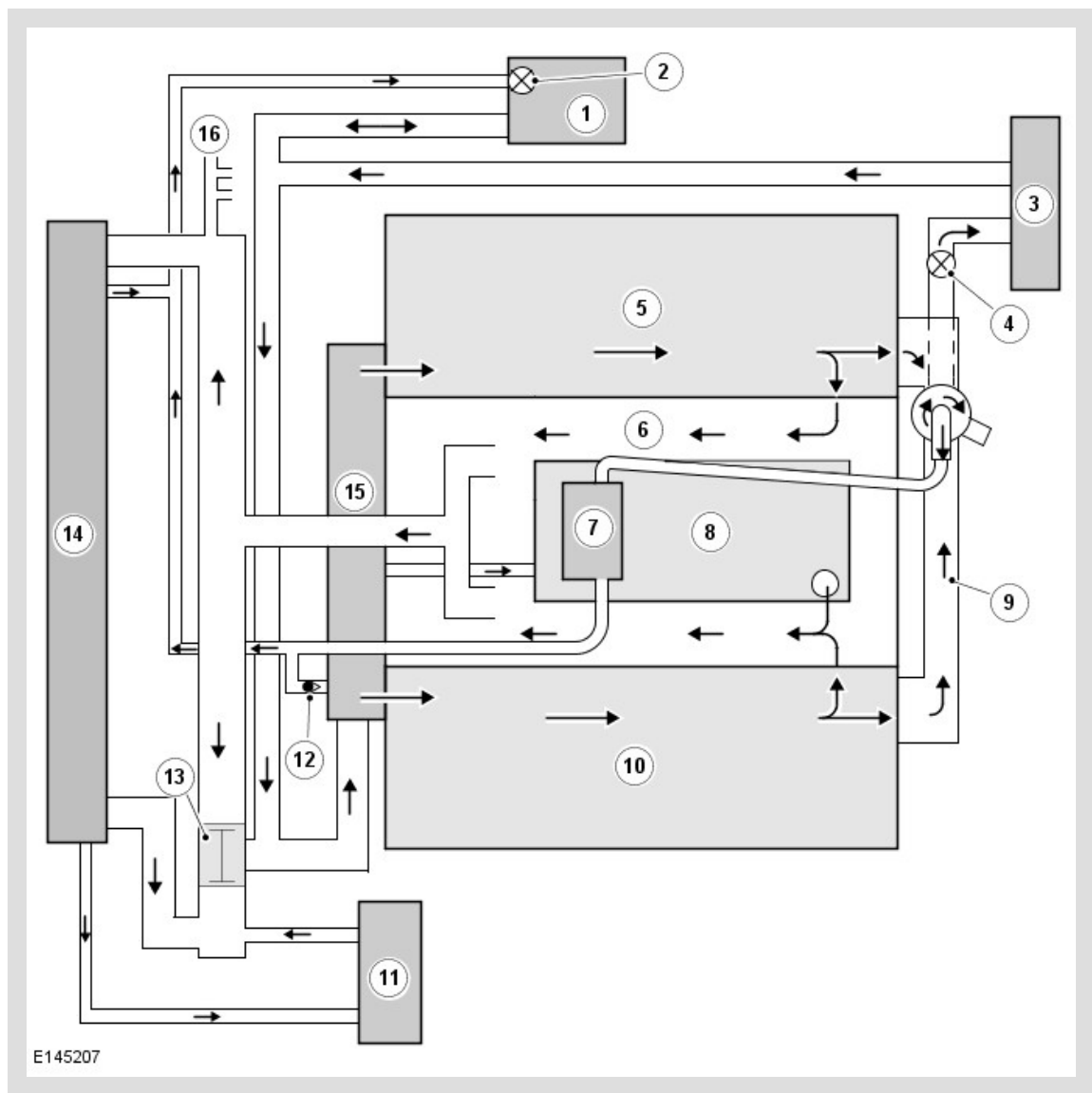
Expansion and contraction of the coolant is accommodated by an air space in the expansion tank and the compliance of the flexible hoses.

Engine Cooling Flow Diagram - Naturally Aspirated Vehicles



ITEM	DESCRIPTION
1	Expansion tank
2	Bleed screw
3	Heater system
4	Bleed screw
5	Right cylinder head
6	Cylinder block
7	Throttle
8	Engine oil cooler
9	Heater manifold
10	Left cylinder head
11	Transmission fluid cooler
12	Check valve
13	Thermostat
14	Radiator
15	Coolant pump

Engine Cooling Flow Diagram - Supercharger Vehicles



ITEM	DESCRIPTION
1	Expansion tank
2	Bleed screw
3	Heater system
4	Bleed screw
5	Right cylinder head
6	Cylinder block
7	Throttle
8	Engine oil cooler
9	Heater manifold
10	Left cylinder head
11	Transmission fluid cooler
12	Check valve

13	Thermostat
14	Radiator
15	Coolant pump
16	Connections with SC cooling system

THERMOSTAT

On both naturally aspirated and supercharger vehicles, the thermostat begins to open at 88 - 90 °C (190 - 194 °F) and is fully open at 102 °C (216 °F).

COOLANT LEVEL SENSOR

If the coolant level in the expansion tank decreases below a predetermined value, the coolant level sensor connects a ground to the CJB (central junction box) , which sends a message to the instrument cluster on the high speed CAN (controller area network) powertrain bus to display a warning message.

For additional information, refer to: Instrument Cluster (413-01 Instrument Cluster, Description and Operation).

ENGINE COOLING FANS

The ECM determines when to operate the cooling fans from the ECT (engine coolant temperature) sensor input and a cooling fan request from the ATC (automatic temperature control) module. The ECM also adjusts the fan speed to compensate for the ram effect of vehicle speed.

VISCOUS COOLING FAN

To control the viscous cooling fan, the ECM sends a PWM signal to the fan control module. The ECM varies the duty cycle of the PWM signal between 0 and 100% to vary the clutch engagement and thus fan speed. If the PWM signal is outside the 0 to 100% range, the fan control module interprets the signal as an open or short circuit and runs the fan at minimum speed to ensure the fan does not overspeed.

If the electrical connections to the viscous fan are disconnected the fan will idle and the engine may overheat. If the ECM detects a cooling fan fault it stores the appropriate DTC (diagnostic trouble code) and signals the instrument cluster on the high speed CAN powertrain bus to display a warning message.

ELECTRIC COOLING FAN

The ECM varies the duty cycle of the PWM signal to the fan control module between 0 and 100% to operate the fan motor in one of four modes: Off, minimum speed (750 rev /min), linear variable speed between minimum and maximum speeds, and maximum speed (2820 rev/min). Under hot operating conditions, the electric cooling fan may continue to operate for up to five minutes after the engine has been switched off.

The fan control module monitors for over and under input voltage, a stalled motor and a partially stalled motor. If it detects any of these faults the fan control module temporarily pulls the PWM signal to ground to notify the fault to the ECM . The length of time the PWM signal is pulled to ground varies between 2 and 8 seconds, depending on the fault. The fan control module repeats the notification process at 5 second intervals until the fault is cleared. If there is more than one fault, only the highest priority fault is notified to the ECM . Fault priorities, in descending order, are over voltage, under voltage, stalled motor and partially stalled motor. The ECM records a related DTC for faults notified by the fan control module and signals the instrument cluster on the high speed CAN powertrain bus to display a warning message.

The nominal operating voltage for the engine cooling fans is 9 to 16 volts. If the input voltage is outside of these limits, the fan control module stops operation of the cooling fan and notifies the ECM of the fault. After an under or over voltage occurs, cooling fan operation is resumed when the input voltage increases to 9.5 volts, or decreases to 15.5 volts, respectively.

To check for a stalled motor, each time it supplies power to start the fan motor the fan control module checks the fan motor speed after 2 seconds. If the speed is zero, the fan control module determines the fan motor is stalled and tries a restart. For a restart, the fan control motor disconnects the power from the fan motor then immediately reconnects it at an increased current level. The fan control module will try up to six restarts of a stalled motor, increasing the output current each time. If the fan motor remains stalled after the sixth restart, the fan control module waits 40 seconds then notifies the ECM of the fault and begins the start process again.

To check for a partially stalled motor, while the fan is running the fan control module monitors the current draw against fan speed. If the current draw is over the limit for more than 10 seconds, the fan control module disconnects the power from the fan motor and notifies the ECM of the fault. After 40 seconds, the fan control module starts the motor again.

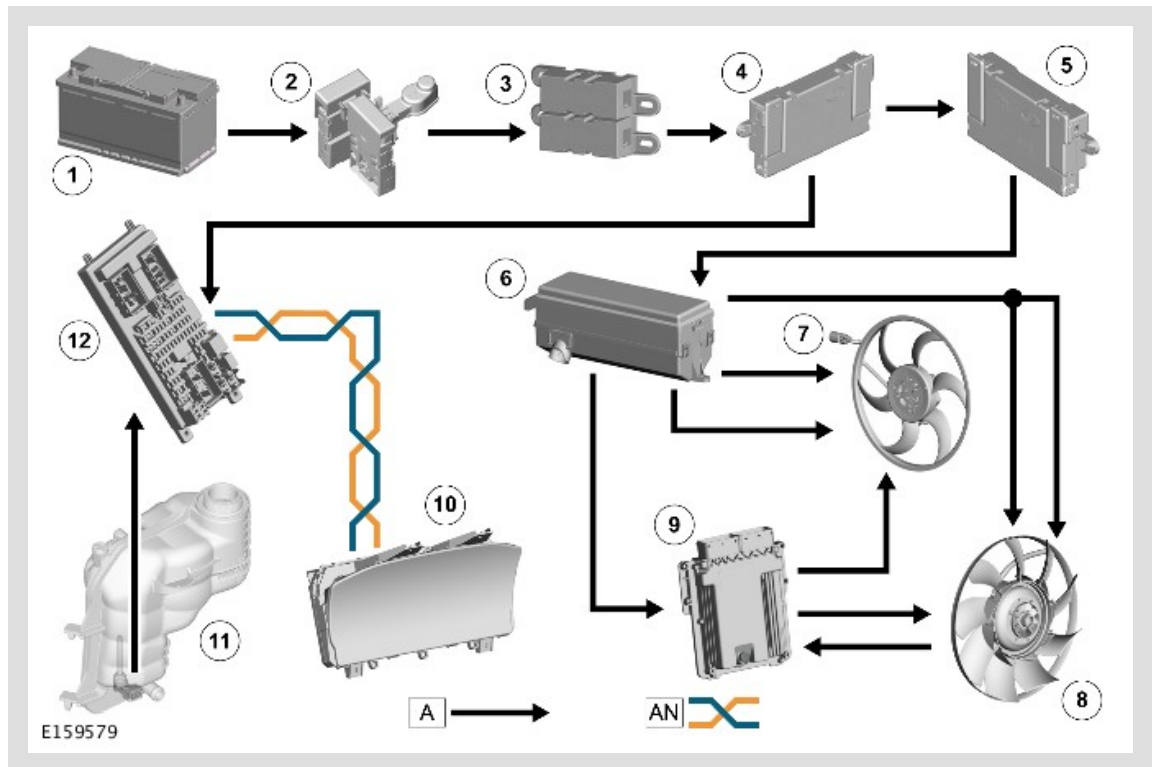
The fan control module incorporates a temperature sensor to prevent damage caused by excessive heat at high ambient temperatures. Operation of the fan control module is discontinued if the temperature reaches 135 °C (275 °F). Operation resumes when the temperature decreases to 120 °C (248 °F).

If a fault occurs with the ignition signal or the PWM signal, the fan control module adopts the following fan speeds:

SIGNAL	FAULT	FAN SPEED	
		IGNITION ON	IGNITION OFF
PWM	Duty cycle implausible	Maximum	Off
PWM	Frequency out of range	Maximum	Off
PWM	Open circuit	Maximum	Off

PWM	Short circuit to battery	Maximum (provided cooling fan is not damaged)	Off
PWM	Short circuit to ground	Maximum (provided cooling fan is not damaged)	Off
Ignition	Open circuit	Off	Off
Ignition	Short circuit to battery	According to PWM duty cycle	Maximum
Ignition	Short circuit to ground	Off (ECM relay fuse may blow)	Off

CONTROL DIAGRAM



A = HARDWIRED; AN = HIGH SPEED CAN POWERTRAIN BUS

ITEM	DESCRIPTION
1	Battery
2	Megafuse
3	Battery Junction Box 2 (BJB2)
4	Battery Junction Box (BJB)
5	Auxiliary Junction Box (AJB)
6	Engine Junction Box (EJB)
7	Electric Cooling Fan
8	Viscous Cooling Fan
9	Engine Control Module (ECM)
10	Instrument Cluster (IC)

11	Coolant Level Sensor
12	Central Junction Box (CJB)